



PROBLEM SOLVING WITH C

UE25CS151B

Prof. Sindhu R Pai, Asst Prof.

Department of Computer Science and Engineering

PROBLEM SOLVING WITH C

Introduction



1. Programming Language(PL)
2. Why PLs? and why so many PLs ?
3. Brief introduction on Paradigms
4. First Programmer
5. Why C after Python?
6. Applications of C
7. History of PLs and Development of C
8. Features of C

What is Programming Language(PL)?

- Formal computer language or constructed language designed to communicate instructions to a machine, particularly a computer (wiki defn)
- A way of telling a machine what operations to perform
- Can be used to create programs to control the behavior of a machine or to express algorithms

Why PLs and why so many PLs?

- To choose the right language for a given problem
- Applications
 - Web Browsers, Social Networks, Image Viewer
 - Facebook.com, Bing, Google.com, Games
 - Various OS

Paradigm – Style of Programming

- Imperative
 - First do this and next do that
 - FORTRAN, Algol, COBOL, Pascal
- Structured programming
 - Single entry and single exit. Avoid GOTO
 - C
- Procedural
 - Subset of imperative
 - Use of subroutines
 - C
- Declarative
 - Declares a set of rules about what outputs should result from which inputs
 - Lisp, SQL

Paradigm – Continued..

- Functional
 - Function should be the first class citizen. No side effects. Assign Function name to another, pass function name as an argument, return function itself
 - If and recursion. No loops
 - Scheme, Haskell, Miranda and JavaScript
- Logical
 - Uses predicates
 - Extraction of knowledge from basic facts and relations
 - ASP, Prolog and Datalog
- Object-Oriented
 - User perception is taken into account.
 - Data needs protection
 - Java, c++, Python

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First Programmer – Ada Lovelace



Why C after Python ?

- Paradigms - Procedural vs OOP
- Aged compared to Python
- Growth rate of PLs - TIOBE Index

Best Applications of C/C++

- ❖ Linux Kernel: It is written in C
- ❖ Adobe Systems: Includes Photoshop and Illustrator
- ❖ Mozilla: Internet Browser Firefox Uses C++.
- ❖ Bloomberg: Provides real time financial information to investors
- ❖ Callas Software: Supports PDF creation, optimization, updation tools and plugins
- ❖ Symbian OS: Used in cellular phones.

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History of PLs

- 1820-1850 England, Charles Babbage invented two mechanical Computational device i.e., Analytical Engine and Difference Engine
- In 1942, United States, ENIAC used electrical signals instead of physical motion
- In 1945, Von Neumann developed two concepts: Shared program technique and Conditional control transfer
- In 1949, Short code appeared
- In 1951, Grace Hopper wrote first compiler, A-0
- Fortran: 1957, John Backus designed.
- Lisp, Algol- 1958
- Cobol: 1959
- Pascal: 1968, Niklaus Wirth
- C: 1972, D Ritchie
- C++: 1983, Bjarne Stroustrup, Compile time type checking, templates are used.
- Java: 1995, J. Gosling, Rich set of APIs and portable across platform through the use of JVM

History.. Continued...

- Development of C
 - Martin Richards, around 60's developed BCPL [Basic Combined Programming Language]
 - Enhanced by Ken Thompson and Introduced B language.
 - C is Originally developed between 1969 and 1973 at Bell Labs by Dennis Ritchie and Kernighan. Closely tied to the development of the Unix operating system
- Standardized by the ANSI [American National Standards Institute] since 1989 and subsequently by ISO [International Organization for Standardization].
 - C89, C99, C11

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FYK

- Charles Babbage – Father of Computer



- Dennis Ritchie



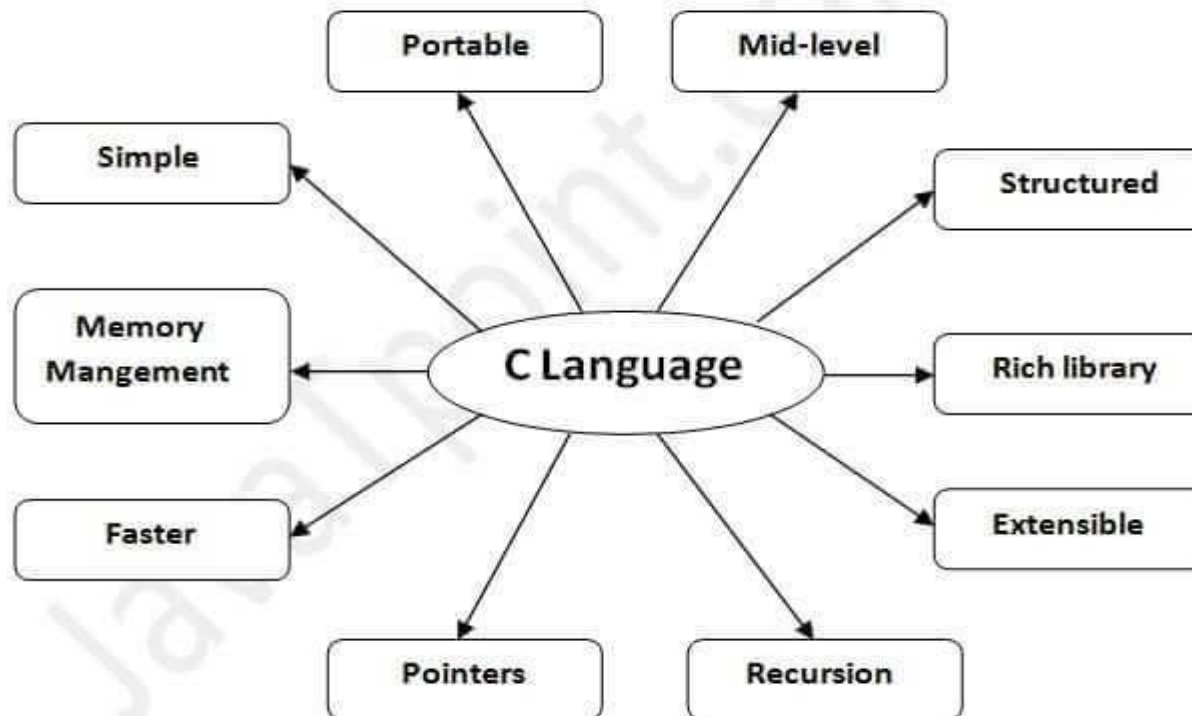
- Brian Kernighan



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Features of C





THANK YOU

Dr. Shylaja S S, Director, CCBD & CDSAML, PESU

Prof. Sindhu R Pai

sindhurpai@pes.edu

Department of Computer Science and Engineering

Ack: Mr. U Shivakumar

Teaching Assistant, CSE, PES University